

```
import pandas as pd
```

```
df = pd.read_csv('amazon.csv')
```

```
df.head()
```

	reviewText	Positive
0	This is a one of the best apps according to a b...	1
1	This is a pretty good version of the game for ...	1
2	this is a really cool game. there are a bunch ...	1
3	This is a silly game and can be frustrating, b...	1
4	This is a terrific game on any pad. Hrs of fun...	1

```
df.shape
```

```
(20000, 2)
```

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
nltk.download("vader_lexicon")
nltk.download("stopwords")
nltk.download("wordnet")
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package vader_lexicon to /root/nltk_data...
[nltk_data] Package vader_lexicon is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
True
```

```
df.head()
```

	reviewText	Positive
0	This is a one of the best apps according to a b...	1
1	This is a pretty good version of the game for ...	1
2	this is a really cool game. there are a bunch ...	1
3	This is a silly game and can be frustrating, b...	1
4	This is a terrific game on any pad. Hrs of fun...	1

```
from nltk.sentiment.vader import SentimentIntensityAnalyzer
```

```
analyzer = SentimentIntensityAnalyzer()
```

```
def get_sentiment(text):
    scores = analyzer.polarity_scores(text)
    sentiment = 1 if scores["pos"] > 0 else 0
    return sentiment
```

```
df["sentiment"] = df["reviewText"].apply(get_sentiment)
```

```
df.head()
```

	reviewText	Positive	sentiment
0	This is a one of the best apps according to a b...	1	1
1	This is a pretty good version of the game for ...	1	1
2	this is a really cool game. there are a bunch ...	1	1
3	This is a silly game and can be frustrating, b...	1	1
4	This is a terrific game on any pad. Hrs of fun...	1	1

```
scores = analyzer.polarity_scores("one best apps according bunch people agree bomb")
```

```
scores
```

```
{'neg': 0.215, 'neu': 0.336, 'pos': 0.45, 'compound': 0.5423}
```

```
from sklearn.metrics import confusion_matrix
result = confusion_matrix(df["Positive"],df["sentiment"])
result
```

```
array([[ 1377,  3390],
       [   620, 14613]])
```

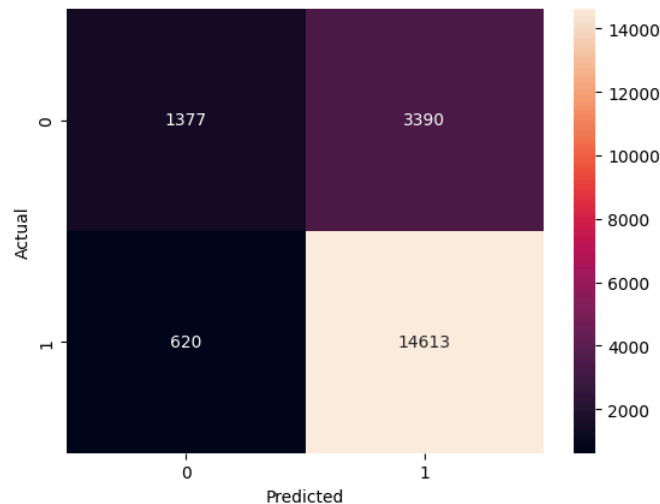
```
from sklearn.metrics import classification_report
print(classification_report(df["Positive"],df["sentiment"]))
```

	precision	recall	f1-score	support
0	0.69	0.29	0.41	4767
1	0.81	0.96	0.88	15233
accuracy			0.80	20000
macro avg	0.75	0.62	0.64	20000
weighted avg	0.78	0.80	0.77	20000

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
c_mtxrx = pd.crosstab(df["Positive"],df["sentiment"], rownames=["Actual"], colnames=["Predicted"])
c_mtxrx
sns.heatmap(c_mtxrx, annot=True, fmt=".0f")
```

```
<Axes: xlabel='Predicted', ylabel='Actual'>
```



```
count0 = df["Positive"].value_counts()[0]
count1 = df["Positive"].value_counts()[1]
print("Actual Dataset")
print("0's", count0)
print("1's", count1)
```

```
Actual Dataset
0's 4767
1's 15233
```

```
labels = ["0", "1"]
counts = [count0, count1]
plt.figure(figsize=(4,4))
plt.title("Count of 0 and 1 in Training Dataset")
for i, count in enumerate(counts):
    plt.text(i, count, str(count), ha='center', va='bottom', fontsize=10)
plt.bar(labels, counts)
plt.show()
```

```
-----  
NameError                                Traceback (most recent call last)  
/tmp/ipykernel_6187/1403438270.py in <cell line: 0>()  
    1 labels = ["0", "1"]  
----> 2 counts = [count0, count1]  
    3 plt.figure(figsize=(4,4))  
    4 plt.title("Count of 0 and 1 in Training Dataset")  
    5 for i, count in enumerate(counts):  
  
NameError: name 'count0' is not defined
```

Start coding or [generate](#) with AI.